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Problem

Let N be a positive integer. Initially, the numbers from 1 to N are written on a board in increasing order. In each step, Facu chooses two numbers a and b currently on the board and replaces them either with $a+k$ and $b-k$ or with $a-k$ and $b+k$, where k is the number of numbers on the board strictly between a and b. For example, if at some point the numbers on the board are

$$1,2,5,6,4,3,$$

Facu could choose 2 and 6 and replace them either with 3 and 5 or with 1 and 7 , so that the board becomes

$$1,3,5,5,4,3 \quad \text{or} \quad 1,1,5,7,4,3,$$

respectively. Determine all positive integers N for which Facu can, after a sequence of steps, arrange the numbers on the board in decreasing order from N down to 1.

Solution. We claim that Facu can arrange the numbers on the board in decreasing order from N down to 1 if and only if $N \equiv \{0,1,3\} \pmod 4$.

Claim 1 — The sum of all the numbers in the even positions is invariant.

Proof. Say a and b are both in the even positions. Then obviously the sum remains invariant as we add and subtract k.
If a and b are in opposite positions (wolog a is in the even position), then k is an even number. As a result, if we add or subtract k, in either case the sum changes by $\pm k \equiv 0 \pmod 2$. □

Suppose $N = 4k+2$ for some $k \in \mathbb{Z}$ then, initially the sum of the numbers in even positions is

$$2+4+\cdots+(4k+2) \equiv (2k+1)(2k+2) \equiv 0 \pmod 2.$$

Finally, the sum of the numbers in even positions is

$$(4k+1)+(4k-1)+\cdots+3+1 \equiv (2k+1)^2 \equiv 1 \pmod 2.$$

This is a contradiction. As a result $N \not\equiv 2 \pmod 4$.

Say $N = 2k+1$ for some $k \in \mathbb{Z}$. We will perform induction.
For $N = 3$, we do the following

$$(1,2,3) \to (2,2,2) \to (3,2,1).$$

Now assume that we can do this for $N = 2n+1$ where $n \in \{1,2,\dots,k-1\}$. We will show that we can do it for $N = 2k+1$ as well.
We choose $a = 1, b = 2k+1$. As a result we get,

$$(1,2,3,\dots,2k+1) \to (2k,2,3,\dots,2k,2).$$

Let a_i denote the number at the ith position.
We will do the following transformations: $(a_{2i-1}+1, a_{2i+1}-1)$ for $i \in \{1,2,\dots,k\}$ in order (starting from 1).
Observe that we have succesfully transformed

$$(2k,2,3,\dots,2k,2) \to (2k+1,2,3,\dots,2k,1).$$

Now we since we have $2,3,\dots,2k$ which are $2k-1$ many numbers, by induction we can transform it to $2k, 2k-1, \dots, 2$ since 1 and $2k+1$ will never occur "between" two numbers.

Say that $N = 4k$ for some $k \in \mathbb{Z}$. We will perform induction.
For $N = 4$ we do the following

$$(1,2,3,4) \to (3,2,3,2) \to (4,2,2,2) \to (4,3,2,1).$$

Now assume that we can do this for $N = 4n$ where $n \in \{1,2,\dots,k-1\}$. We will show that we can do it for $N = 4k$ as well.
We do the following transformation

$$(1,2,3,\dots,4k-1,4k) \to (4k-1,2,3,\dots,4n-1,2).$$

We will do the following transformations: $(a_{2i-1}+1, a_{2i+1}-1)$ for $i \in \{1,2,\dots,2n-1\}$ in order (starting from 1).
Then, we will do the following transformations: $(a_{2i}+1, a_{2i+2}-1)$ for $i \in \{1,2,\dots,2n-1\}$ in order (starting from $2n-1$).
Observe that we finally get

$$(4k,3,3,4,\dots,4k-1,4k-1,1).$$

Now we do, $(a_2 + 4k - 4, a_{4k-1} - (4k-4))$ to get

$$(4k,4k-1,3,4,\dots,4k-2,2,1).$$

Now by induction, since we have $3, 4, \dots, 4k-2$ (which are $4(k-1)$ in number), we can rearrange this in decreasing order since $4n, 4n-1, 2, 1$ will never occur "between" two numbers.

As a result, Facu can arrange the numbers on the board in decreasing order from N down to 1 if and only if $N \equiv \{0, 1, 3\} \pmod 4$. $\square$